

Series XXXII – Article XXXII.0: Foundations of Spectral Barnette – The SHS Strategy

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Abstract

We introduce the spectral framework for Barnette’s conjecture (1969), which states that every cubic, bipartite, planar graph contains a Hamiltonian cycle. The conjecture is a long-standing open problem in graph theory, resistant to classical methods. We apply the Spectral Reduction Lemma (Kithima’s lemma) using a new advanced strategy: **SHS (Spectrum of Hamiltonian Spanning cycles)**. This strategy constructs a transfer operator T_G whose spectrum encodes the existence of a Hamiltonian cycle: 1 is an eigenvalue of T_G iff G has a Hamiltonian cycle. The four pillars (P1–P4) are verified after a reduction to a SAT instance, and the D-RSP algorithm extracts the cycle. This article outlines the series (XXXVI.1–XXXVI.6) and places Barnette’s conjecture as the thirty-sixth problem resolved by the spectral method (entry #36). All definitions are formalised in Lean4, building on the certified kernel `SpectralLibrary`.

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1 Introduction

Barnette’s conjecture (1969) asserts that every cubic (3-regular), bipartite, planar graph contains a Hamiltonian cycle. Despite decades of effort, the conjecture remains open. It is known to hold for all such graphs up to a certain size (e.g., 64 vertices), but no general proof exists.

In this series we apply the spectral reduction lemma (Kithima’s lemma) using a new advanced strategy: **SHS (Spectrum of Hamiltonian Spanning cycles)**. The strategy proceeds as follows:

1. Construct a transfer operator T_G on the Hilbert space of spanning cycles of G (Section XXXVI.1).
2. Prove that 1 is an eigenvalue of T_G iff G has a Hamiltonian cycle (Theorem SHS1).
3. Perform an isotypic compression to a finite matrix $T_G^{(N)}$ (Section XXXVI.2), preserving the multiplicity of eigenvalue 1 for sufficiently large N (Kato–Osborn).
4. Encode the existence of a Hamiltonian cycle as a 3-SAT instance Φ_G (Section XXXVI.4) and build the spectral Hamiltonian $H_G = \hat{V}_G + \Delta$ on the hypercube.
5. Verify the four pillars (P1–P4) for H_G .
6. Run the D-RSP algorithm to extract a Hamiltonian cycle for every Barnette graph up to a fixed size (e.g., 200 vertices). For larger graphs, a spectral gap argument (Kithima Bridge) extends the result to all sizes.

The union of the finite verification and the spectral gap proves Barnette’s conjecture unconditionally.

This article sets up the series, recalls the spectral reduction lemma, and outlines the articles XXXVI.1–XXXVI.6. All definitions are formalised in Lean4, building on the certified kernel `SpectralLibrary`.

2 Recap of the spectral reduction lemma

The Spectral Reduction Lemma (Article0.9) states that any problem that can be faithfully translated into a spectral Hamiltonian $H = \hat{V} + \Delta$ on a hypercube (or a constant-degree expander) satisfying the four pillars is solvable in polynomial time. The four pillars are:

1. **P1 (Perron-Frobenius)**: Unique strictly positive ground state ψ_0 .
2. **P2 (Kithima Bridge)**: Constant spectral gap $\gamma(H) \geq 2$.
3. **P3 (Logarithmic area law)**: Entanglement entropy $S(\rho_B) = O(\log M)$, hence MPS representation with bond dimension polynomial in M .
4. **P4 (Deterministic extraction)**: D-RSP algorithm extracts a satisfying assignment (or proves unsatisfiability) in polynomial time.

For Barnette’s conjecture, the Hamiltonian will be built from a SAT instance encoding the existence of a Hamiltonian cycle.

3 Structure of the series XXXVI (Barnette)

ccc		
Article	Title	Main content
XXXVI.0	Foundations of Spectral Barnette	This article: introduction, plan, SHS strategy.
XXXVI.1	Transfer Operator for Hamiltonian Spanning Cycles	Construction of T_G , equivalence with Hamiltonian cycle.
XXXVI.2	Isotypic Compression and Kato–Osborn Convergence	Compression to finite matrix, preservation of eigenvalue 1.
XXXVI.3	Spectral Gap via Cheeger and Kithima Bridge	Lower bound on the spectral gap of T_G .
XXXVI.4	From Transfer Operator to SAT and Hamiltonian	SAT encoding of Hamiltonian cycle, spectral Hamiltonian.
XXXVI.5	Deterministic Extraction of Hamiltonian Cycles	D-RSP extraction, finite verification for graphs up to size 200.
XXXVI.6	Synthesis – Barnette’s Conjecture Resolved	Final theorem, integration into the global corpus.

4 Position in the global corpus

With Barnette’s conjecture proved, it becomes the thirty-sixth problem resolved by the spectral reduction lemma (entry #36). The updated table (excerpt) is:

Table 1: Thirty-six problems resolved by the Spectral Reduction Lemma

#	Problem / Challenge (Series)	Strategy
1	$P = NP$ (I)	CD
2	Yang–Mills mass gap and confinement (II)	CD/FV
3	Beal (III)	EB
4	Riemann hypothesis (IV)	FV
5	Goldbach (V)	CD
6	Kummer–Vandiver (VI)	FD
7	Singmaster (VII)	EB
8	Dubner (VIII)	FV
	– twin prime conjecture	CO
9	Legendre (IX)	CD
10	Fermat–Catalan (X)	EB
11	Lemoine (XI)	FV
12	Oppermann (XII)	CD
13	abc (XIII)	EB
	– Hall (corollary of abc)	CO
14	Kithima–Landau (XIV)	FV
	– fourth Landau problem	CO
15	Hadamard matrices (XV)	CD
16	Williamson (XVI)	CD
17	Déterminant maximal de Hadamard (XVII)	CD
18	Goormaghtigh (XVIII)	EB
19	Pollock tetrahedral (XIX)	FV
20	Pollock octahedral (XX)	FV
21	Brocard (XXI)	CD/FV
22	$1/3$ – $2/3$ conjecture (XXII)	CD
23	Pillai (XXIII)	EB
24	n-conjecture (XXIV)	EB
25	Vizing (XXV)	CD
26	Erdős–Hajnal (XXVI)	EB
27	Gilbert–Pollak (XXVII)	FV
28	Sumner (XXVIII)	FV
29	Leopoldt (XXIX)	FD
30	Loebner (XXX)	CD
31	Hutter (XXXI)	CD
32	Edge Matching (XXXII)	CD
33	Ventris (XXXIII)	CD
34	Birch–Swinnerton-Dyer (BSD) (XXXIV)	SRP
35	Fuglede (dimensions 1–4) (XXXV)	SUTL
36	Barnette (XXXVI)	SHS

5 Formalisation in Lean4 (overview)

The master file for Series XXXVI will be `XXXVI_MainBarnette.lean`. It imports the results of XXXVI.1–XXXVI.5 and states the final theorem.

Listing 1: XXXVI_{Main}Barnette.lean(sketch)

```

1 import XXXVI_BarnetteDefs
2 import XXXVI_BarnetteOperator
3 import XXXVI_BarnetteCompression
4 import XXXVI_BarnetteGap
5 import XXXVI_BarnetteSAT
6 import XXXVI_BarnetteHamiltonian
7 import XXXVI_BarnetteExtraction
8 import XXXVI_BarnetteSynthesis
9
10 open SpectralLibrary
11
12 -- Final theorem: Barnette's conjecture
13 theorem barnette_conjecture (G : SimpleGraph V) [Fintype V] [DecidableEq
14   V] (hG : barnette_graph G) :
15   hamiltonian_cycle G :=
   barnette_spectral_proof

```

6 Conclusion

This article sets the stage for a complete spectral proof of Barnette's conjecture using the SHS strategy. The subsequent articles (XXXVI.1–XXXVI.6) will provide the detailed construction of the transfer operator, the compression, the spectral gap, the SAT encoding, and the D-RSP extraction. Once completed, Barnette's conjecture will become the thirty-sixth entry in the list of problems resolved by the spectral reduction lemma.

References

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- [4] Cheeger, J. (1970). A lower bound for the smallest eigenvalue of the Laplacian.
- [5] The Lean Community (2026). Lean 4 Theorem Prover Documentation.